FDS EXPERIMENT- 1

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Exp No:**1.a** Analyze the trend of data science job postings over the last decade

Description: Use web scraping (e.g., BeautifulSoup) or APIs (e.g., LinkedIn API) to gather data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib/seaborn for visualization.

```
[ ]
import pandas as pd
import matplotlib.pyplot as plt

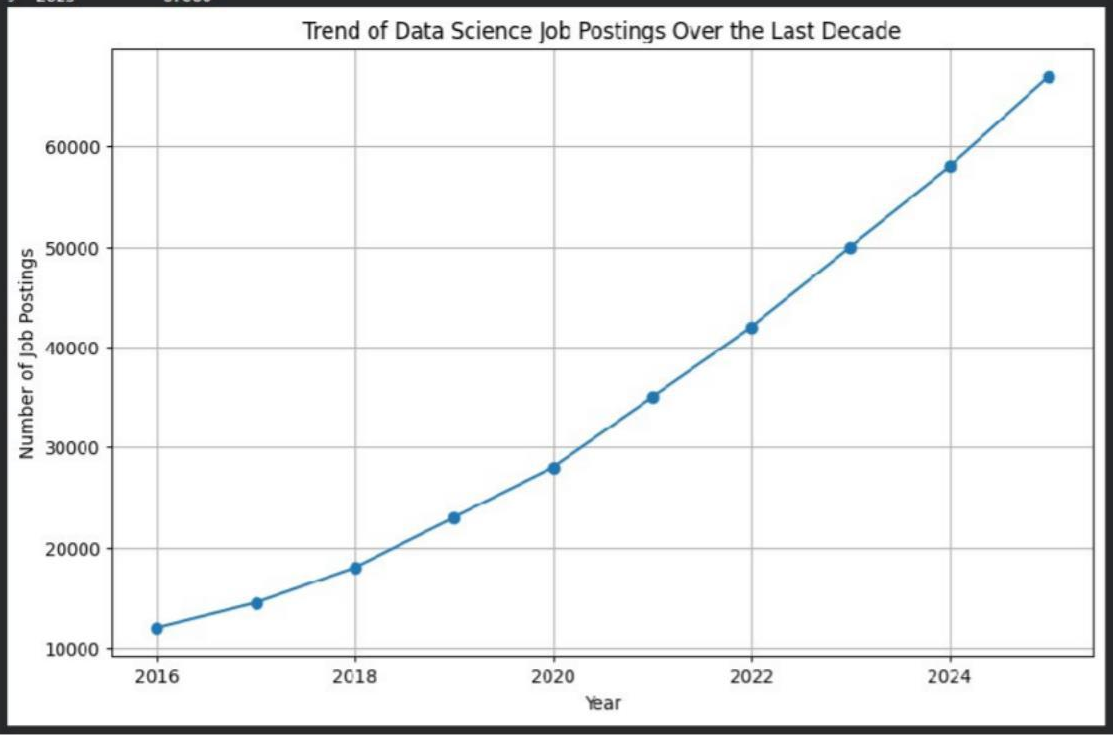
years = list(range(2016, 2026))
job_postings = [12000, 14500, 18000, 23000, 28000, 35000, 42000, 50000, 58000, 67000]

df = pd.DataFrame({
    'Year': years,
    'Job_Postings': job_postings
})

print(df)

plt.figure(figsize=(10, 6))
plt.plot(df['Year'], df['Job_Postings'], marker='o')
plt.title('Trend of Data Science Job Postings Over the Last Decade')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.grid(True)
plt.show()
```

```
   Year  Job_Postings
0  2016         12000
1  2017         14500
2  2018         18000
3  2019         23000
4  2020         28000
5  2021         35000
6  2022         42000
7  2023         50000
8  2024         58000
9  2025         67000
```



Exp No:**1.b** Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

Description: Use a dataset of job postings and categorize them into different roles. Visualize the distribution using pie charts or bar plots.

```python
import pandas as pd
import matplotlib.pyplot as plt

data = {
    'Role': [
        'Data Analyst', 'Data Scientist', 'Data Engineer',
        'Data Analyst', 'Data Scientist', 'Data Engineer',
        'Data Analyst', 'Machine Learning Engineer',
        'Data Scientist', 'Data Engineer', 'Data Analyst',
        'Machine Learning Engineer', 'Data Scientist', 'Data Analyst',
        'Data Engineer'
    ]
}

df = pd.DataFrame(data)

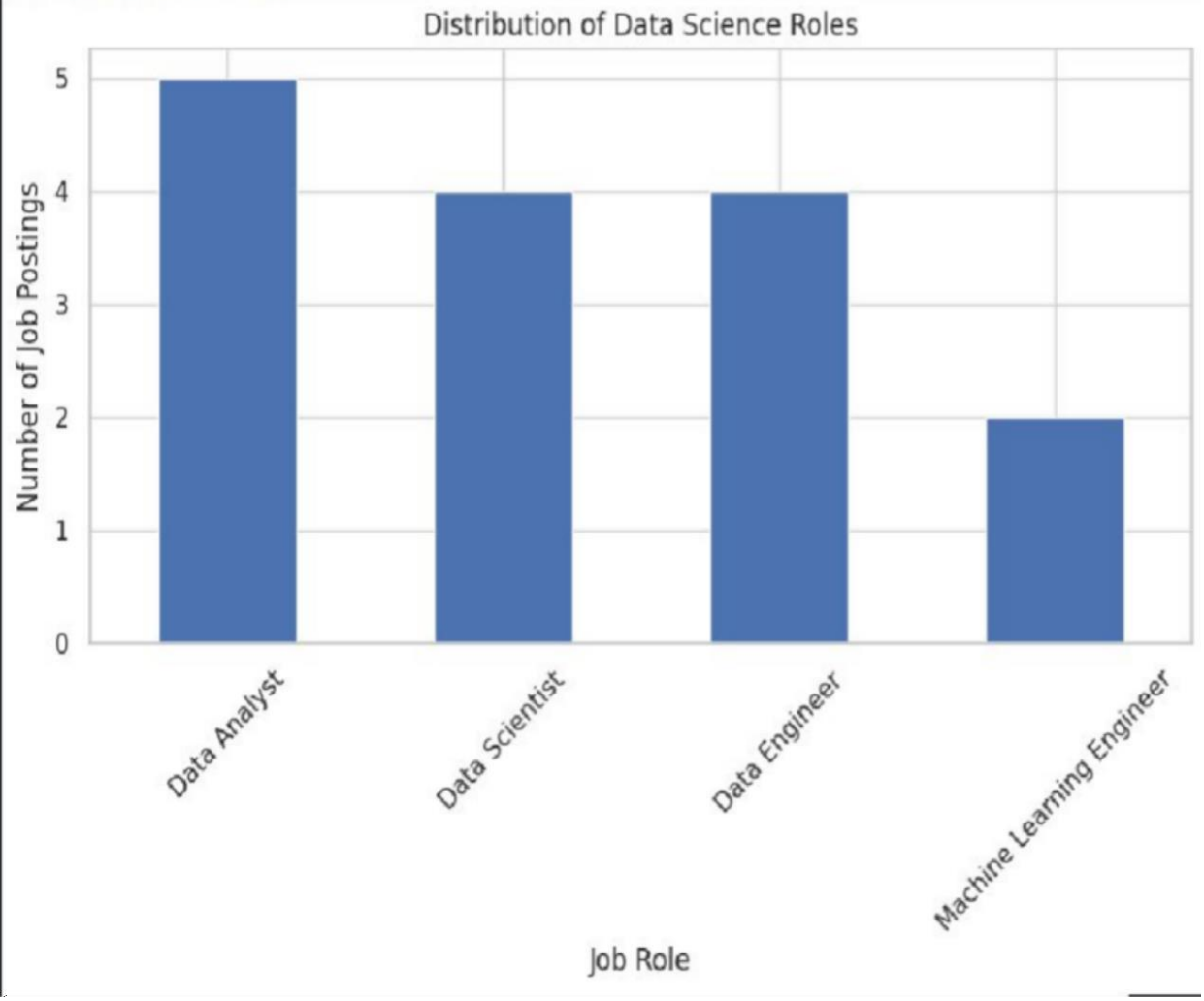
role_counts = df['Role'].value_counts()

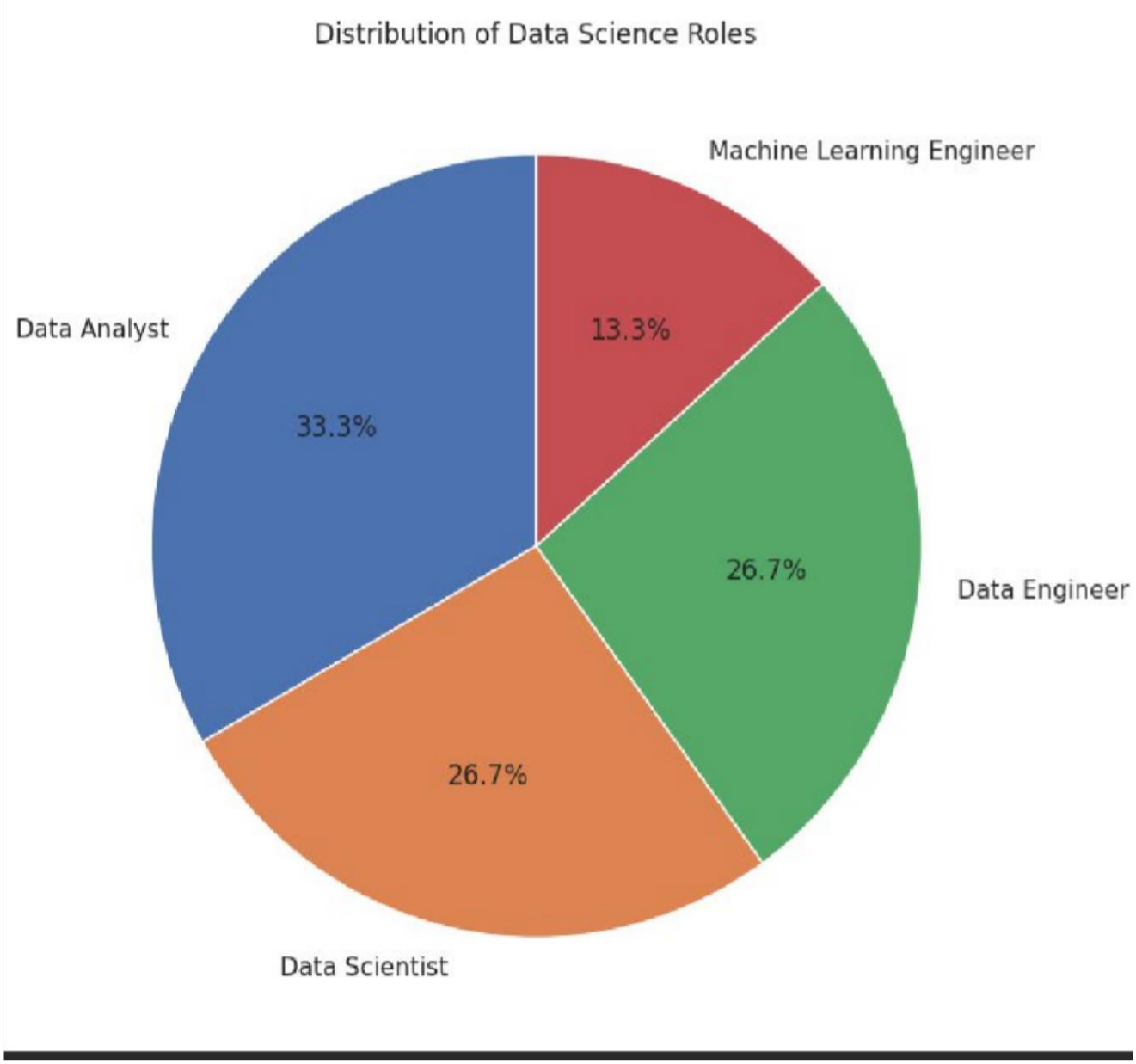
print(role_counts)

plt.figure(figsize=(8, 6))
role_counts.plot(kind='bar')
plt.title('Distribution of Data Science Roles')
plt.xlabel('Job Role')
plt.ylabel('Number of Job Postings')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 8))
role_counts.plot(kind='pie', autopct='%1.1f%%', startangle=90)
plt.title('Distribution of Data Science Roles')
plt.ylabel('')
plt.show()
```

```
Role
Data Analyst                5
Data Scientist              4
Data Engineer               4
Machine Learning Engineer   2
Name: count, dtype: int64
```




Distribution of Data Science Roles
Machine Learning Engineer
13.3%
Data Analyst
33.3%
26.7%
Data Engineer
26.7%
Data Scientist

Exp No:**1.c** Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

Description: Create small datasets for each type and explain their characteristics.

```
import pandas as pd
import json

structured_data = pd.DataFrame({
    'ID': [101, 102, 103],
    'Name': ['Arun', 'Bala', 'Kumar'],
    'Age': [20, 21, 19]
})

print("STRUCTURED DATA")
print(structured_data)
print("\nCharacteristics: Rows and columns, fixed structure, easy to analyze.")


semi_structured_data = [
    {"ID": 101, "Name": "Arun", "Department": "CSE"},
    {"ID": 102, "Name": "Bala", "Department": "ECE"}
]

print("\nSEMI-STRUCTURED DATA")
print(json.dumps(semi_structured_data, indent=2))
print("\nCharacteristics: Uses keys/tags, flexible structure, example: JSON.")


unstructured_data = [
    "Arun is a CSE student.",
    "Bala likes Python."
]

print("\nUNSTRUCTURED DATA")
print(unstructured_data)
print("\nCharacteristics: No fixed structure, example: text, images, audio and video.")
```

```
STRUCTURED DATA
    ID   Name  Age
0  101   Arun   20
1  102   Bala   21
2  103  Kumar   19

Characteristics: Rows and columns, fixed structure, easy to analyze.

SEMI-STRUCTURED DATA
[
  {
    "ID": 101,
    "Name": "Arun",
    "Department": "CSE"
  },
  {
    "ID": 102,
    "Name": "Bala",
    "Department": "ECE"
  }
]

Characteristics: Uses keys/tags, flexible structure, example: JSON.

UNSTRUCTURED DATA
['Arun is a CSE student.', 'Bala likes Python.']

Characteristics: No fixed structure, example: text, images, audio and video.
```

Exp No:**1.d** Conduct an experiment to encrypt and decrypt given sensitive data.

Description: Use the cryptography library to encrypt and decrypt a piece of data.

```
!pip install cryptography

from cryptography.fernet import Fernet

key = Fernet.generate_key()

f = Fernet(key)

token = f.encrypt(b"Rajalakshmi Engineering College")

print("Original Data:")
print(b"Rajalakshmi Engineering College")

print("\nEncryption Key:")
print(key)

print("\nEncrypted Data:")
print(token)

decrypted_data = f.decrypt(token)

print("\nDecrypted Data:")
print(decrypted_data)
```

```
Requirement already satisfied: cryptography in /usr/local/lib/python3.12/dist-packages (43.0.0)
Requirement already satisfied: cffi>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from cryptography) (2.1.0)
Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-packages (from cffi>=2.0.0->cryptography) (3.0)
Original Data:
b'Rajalakshmi Engineering College'

Encryption Key:
b'ceSIhD0FgJzut5HtqUGY1-qmDCjBygYxbCp6UaRcSUI='

Encrypted Data:
b'gAAAABqgGWWfo1-4RzXs8Qm00y7R54jjBFEBA3fGXaYTbsH204367_BBzmKS9gpdOepJHIcP4vIaWQjBGZp9QyKSLYhgKbKH_FwRkx9uQqD6MXlVECVG54-'

Decrypted Data:
b'Rajalakshmi Engineering College'
```